From Switches to Systems

The Factory as Part of the AI Architecture

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Networking Purpose-Built for AI Clusters

The network rack has become infrastructure

TODAY	FUTURE
51.2T	102.4T and beyond
100s of fibers	1000s of fibers
Air cooling	Liquid cooling
TOR Integrated	Dedicated network rack



Turning Technology Into a Repeatable Model

These engagements force product, rack and factory decisions to converge around one customer outcome.

1

FABRIC INTENT

Translate topology, traffic and optical requirements

2

SYSTEM CO-DESIGN

Align switch, cooling, power, rack and fiber paths

3

VALIDATE

Signal integrity, leak detection, regulatory and rack test

4

INDUSTRIALIZE

Repeatable factory build, service and deployment model

CUSTOMER PROOF POINT

One joint team resolving optical, thermal, mechanical and manufacturing interdependencies before deployment.

Factory enablement priorities for AI Infrastructure

Large rack integration: ten priorities, grouped into four themes

Physical Systems

Build the rack as one system

01 Rack-level integration
FAT · L11 · cluster validation

02 Liquid cooling
CDUs · QDs · leak testing · leak detection

03 Fiber and cabling
Cleaning · bend radius · routing · labeling

06 Power architecture
ORv3 · 48V

Test & Repair

Prove it works, and fix it fast

04 Fast repair strategy
Rack twins · hot-swap

08 Manufacturing test evolution
Optical · thermal · cluster

Smart Factory & Data

Inspect, predict, and trace

05 Smart factory and AI
Vision · AOI · predictive

10 Digital traceability
Genealogy · MES integration

Scale & People

Replicate at volume

07 Scale readiness
Hundreds of racks a week

09 Workforce development
Cooling certs · optics · safety

Challenges Faced

SWITCH STORAGE CONDITIONS

Custom cold plate designs drive varying N2 decay rates during storage

- Behavior of switches stored with coolant and N2 in place is not yet defined
- Requires coordination across coolant suppliers and cold plate vendors

Direct fulfillment and service depots add further storage and handling variables

FIBER CABLE QUALITY

QA/QC at the cable vendor

Incoming inspection at the factory

Small particulates drive insertion loss, so clean cable is essential

Explicit cleaning requirements and protocols still need to be defined

Is 100% cable cleaning necessary prior to final assembly? Dry vs wet cleaning

Lower quality cables will lead 100% cable inspection prior to final assembly, significantly slowing down the process

Troubleshooting is difficult once cables are bundled

FIBER HANDLING & ROUTING

Operator Training

Standard BICSI training is overly in-depth and cost-prohibitive at factory assembly scale

Routing space within the rack

Maintaining bend radius

Labeling for port identification

Cable bundling

Build sequence management and SOPs

Serviceability

Enables: Ramping New Architectures Quickly

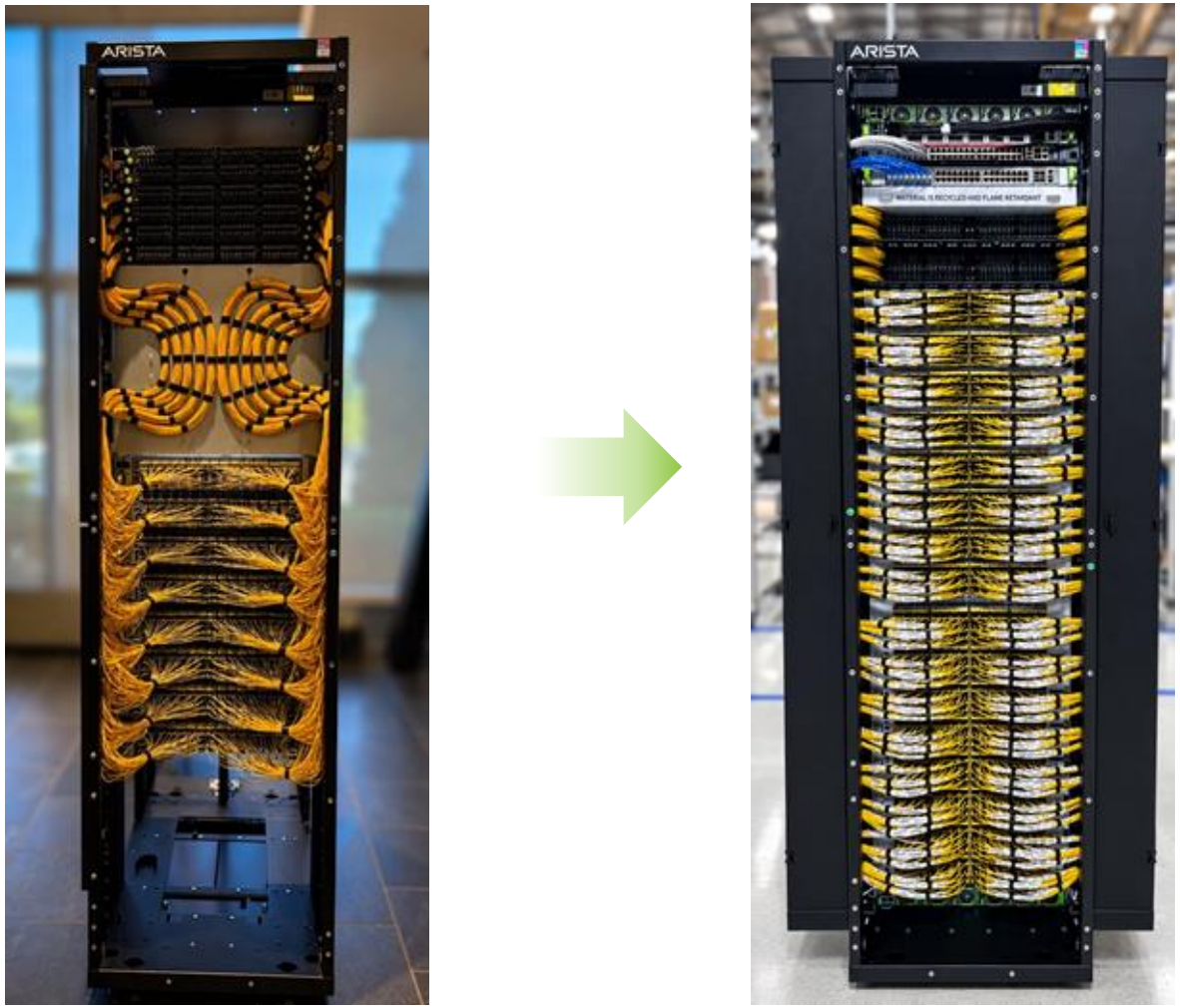
Standardization wherever possible
One manifold, common LC and power infrastructure across racks

Concurrent design and industrialization
DFM/DFA and L11 BOM developed with the switch architecture, gated first article to mass production

Reusable factory assets
Configurable jigs and a standard line layout a new architecture drops into

Test methods built with the architecture
Optical, thermal and cluster test defined during design, not at ramp

Supplier readiness ahead of demand
Long-lead equipment and manifolds engaged early, second sources identified



One Accountable System, Not a Chain of Handoffs

Arista + Jabil differentiates on completeness and joint accountability.

CAPABILITY STACK

Network product + NOS	Arista architecture, EOS / SONiC options
Thermal system	Switch + optics cooling, Mikros, rack loop
Optical path	Transceivers, patching, fiber routing, serviceability
Rack infrastructure	Power, busbar, manifolds, mechanics, controls
Industrialization	Factory build, validation, rack test, deployment

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THE DIFFERENTIATION

Thank You!



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